

DETERMINISTIC PATH ENSEMBLES, DRIVEN COLLISION MAPS, AND ENDOGENOUS CONJUGATE PARAMETERS

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ABSTRACT. We construct a biseam graph completion retaining both vertical and horizontal oriented germs, so the four-port affine return is genuinely bijective on every corner stratum. An explicit autonomous extended Hamiltonian realizes the return with exact unit clock and full-port work. A polyhedral anisotropic current scale, including face, corner, and mixed jets, yields arbitrary finite-order physical response without importing an isotropic Holder spectral gap. Mechanical and valuation parameters coincide only under the stated likelihood-cocycle compatibility axiom.

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1. ROUND-SEVEN RECONSTRUCTION: BISEAM COMPLETION, ANISOTROPIC RESPONSE, AND AN EXPLICIT AUTONOMOUS SUSPENSION

This section is self-contained and replaces the round-six controlling module. The point of the reconstruction is that the return section remembers both the next vertical port and the preceding horizontal port. Thus no two boundary germs are collapsed. The response theory is built on an anisotropic scale for the invertible baker map rather than on an isotropic Holder space.

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1.1. The biseam graph completion. Let

$$0 = s_0(a) < s_1(a) < \cdots < s_4(a) = 1, \quad p_i(a) = s_i(a) - s_{i-1}(a) > 0,$$

and let

$$0 = t_0(a) < t_1(a) < \cdots < t_4(a) = 1, \quad t_i(a) - t_{i-1}(a) = p_i(a).$$

Write $I_i = (s_{i-1}, s_i)$ and $J_k = (t_{k-1}, t_k)$. The *biseam completion* is the finite cell complex

$$\widehat{\Sigma}_a = \coprod_{1 \leq i, k \leq 4} \overline{I_i \times J_k}^g / \sim,$$

where two points are identified only when their oriented germs agree in both coordinate directions. Consequently an interior point has one cell label, a point on one seam has two oriented copies, and a crossing of a vertical and a horizontal seam has four quadrant copies. We write a point as $[q, p; i, k]$ whenever its vertical and horizontal germs belong to I_i and J_k .

On a point with vertical label i define

$$Q_i(q) = \frac{q - s_{i-1}}{p_i}, \quad P_i(p) = t_{i-1} + p_i p.$$

The next vertical germ j is the unique oriented germ for which $Q_i(q) \in \overline{I_j}^g$; the next horizontal germ is i , because $P_i(p) \in \overline{J_i}^g$. Set

$$F_a[q, p; i, k] = [Q_i(q), P_i(p); j, i].$$

Lemma 1.1 (Biseam bijection). *The map $F_a : \widehat{\Sigma}_a \rightarrow \widehat{\Sigma}_a$ is a bijective stratum-preserving exact symplectic map. On every closed cell it is affine, its determinant is one, and its inverse is*

$$q = s_{i-1} + p_i Q, \quad p = \frac{P - t_{i-1}}{p_i},$$

where the horizontal germ of the target uniquely supplies the old branch i . The family is C^{m+2} in a moving-cell charts whenever the functions s_i are C^{m+2} and $p_i \geq p_* > 0$.

Proof. On the source cell (i, k) the displayed affine map has derivative $\text{diag}(p_i^{-1}, p_i)$ and hence preserves $dq \wedge dp$. The target horizontal germ equals i even when $P_i(p)$ lies on a horizontal seam; this is the datum missing from a one-seam completion. It therefore selects one inverse branch. The inverse formula recovers q, p , and the recovered value of p selects the old horizontal germ k , including its oriented side at a seam. Thus both compositions are the identity on every quadrant germ. Since the affine branch primitives differ by exact constants, the maps paste to an exact symplectic map on the finite cell complex. In the coordinates

$$u = (q - s_{i-1})/p_i, \quad v = (p - t_{k-1})/p_k,$$

all incidence maps are fixed and the coefficients are C^{m+2} in a . □

1.2. A complete polyhedral current scale. Let $\mathcal{C}_r$ be the set of codimension- r oriented cells of $\widehat{\Sigma}_a$, $0 \leq r \leq 2$. For a multi-index α tangential to a cell and a normal multi-index β , let $\text{tr}_C^{\alpha,\beta} u$ denote the corresponding current trace. The incidence signs are those of the cellular boundary operator, so corner traces occur as boundaries of face traces and not as independent formal symbols.

For integers $m \geq 0$, $r_u \geq 2m + 4$, and $r_s \geq 2m + 4$, define the strong and weak anisotropic norms by completing cellwise smooth functions for

$$\begin{aligned} \|u\|_{m,+} &= \sum_{C \in \mathcal{C}_0} \|u_C\|_{\mathbf{B}^{r_u, -r_s}} + \sum_{\substack{C \in \mathcal{C}_1 \cup \mathcal{C}_2 \\ |\alpha| + |\beta| \leq m}} \omega_C \|\text{tr}_C^{\alpha,\beta} u\|_{\mathbf{B}^{r_u - |\alpha|, -r_s - |\beta|}}, \\ \|u\|_{m,-} &= \sum_{C \in \mathcal{C}_0} \|u_C\|_{\mathbf{B}^{r_u - 1, -r_s - 1}} + \sum_{\substack{C \in \mathcal{C}_1 \cup \mathcal{C}_2 \\ |\alpha| + |\beta| \leq m}} \omega_C \|\text{tr}_C^{\alpha,\beta} u\|_{\mathbf{B}^{r_u - |\alpha| - 1, -r_s - |\beta| - 1}}. \end{aligned}$$

Here $\mathbf{B}^{r,-s}$ is the completion of Fourier polynomials on a cell with norm

$$\|u\|_{\mathbf{B}^{r,-s}} = \sum_{n, \ell \in \mathbb{Z}} |\widehat{u}(n, \ell)| (1 + |n|)^r (1 + |\ell|)^{-s},$$

using the expanding coordinate in the positive exponent and the stable coordinate in the negative exponent. Fixed partition-of-unity changes give equivalent norms. The weights ω_C are chosen recursively so that an incidence map has norm at most one half of the parent-cell contribution. Denote the two completions by $\mathcal{B}_{a,m}^+$ and $\mathcal{B}_{a,m}^-$. The embedding $\mathcal{B}_{a,m}^+ \hookrightarrow \mathcal{B}_{a,m}^-$ is compact.

Let $\mathcal{L}_a$ be the Perron operator of F_a with respect to Liouville area, acting on all cell and current coordinates by pullback and the cellular incidence rule.

Theorem 1.2 (Anisotropic spectral packet). *For every finite m there are exponents r_u, r_s and constants $C < \infty$, $0 < \rho < 1$, uniform for a in a compact parameter interval, such that*

$$\|\mathcal{L}_a^n u\|_{m,+} \leq C \rho^n \|u\|_{m,+} + C \|u\|_{m,-}.$$

The peripheral spectrum on $\mathcal{B}_{a,m}^+$ consists of the simple eigenvalue one. If $\Pi_a u = \nu_a(u) \mathbf{1}$ and $Q_a = I - \Pi_a$, then

$$S_a = (I - \mathcal{L}_a)^{-1} Q_a = \sum_{n \geq 0} \mathcal{L}_a^n Q_a$$

converges from $\mathcal{B}_{a,m}^+$ to itself. Moreover

$$a \mapsto \mathcal{L}_a \quad \text{is } C^m \text{ as a map } \mathcal{B}_{a,m+j}^+ \rightarrow \mathcal{B}_{a,m}^-$$

for every derivative order $j \leq m$ after the moving-cell identifications.

Proof. On branch i , unstable frequencies are multiplied by p_i^{-1} under the map and stable frequencies are multiplied by p_i under the inverse pullback. Splitting at frequency N and using $p_i \geq p_*$ gives

$$\|\mathcal{L}_a u\|_{B^{r_u, -r_s}} \leq \rho_0 \|u\|_{B^{r_u, -r_s}} + C_N \|u\|_{B^{r_u-1, -r_s-1}}, \quad \rho_0 = \max_i p_i^{\min(r_u, r_s)} < 1,$$

on zero unstable average. Differentiating cell indicators produces their oriented face currents; differentiating those produces the signed corner currents. Because the cellular boundary squares to zero, all multiple incidences cancel in compatible pairs. The recursive choice of ω_C absorbs the finite triangular incidence matrix, and the same estimate holds for every mixed trace of total order at most m .

Compactness follows by a diagonal Fourier cutoff on each of the finitely many cells and traces. Hennion's theorem gives essential spectral radius below one. If $\mathcal{L}_a u = \lambda u$, $|\lambda| = 1$, the unstable Fourier estimate makes u constant along unstable leaves; applying the inverse estimate makes it constant along stable leaves. The biseam Markov graph is the full four-symbol graph, hence u is globally constant. Positivity gives simplicity. The resolvent series follows.

In moving-cell coordinates a parameter derivative is a finite sum of an interior coefficient derivative, a face current, and, after repeated differentiation, a mixed face/corner current. Each derivative consumes one strong derivative and lands in the weak space. The displayed scale contains all such currents through order m , proving the parameter assertion. $\square$

Theorem 1.3 (Finite-order physical response without isotropic-gap input). *Let $G_a \in C^m(\mathcal{B}_{a, 2m+2}^+, \mathbb{R})$. Then $a \mapsto \nu_a(G_a)$ is C^m . Every derivative is a finite sum of typed words in*

$$D^j \mathcal{L}_a, \quad D^j G_a, \quad S_a, \quad \Pi_a,$$

where the domain loss is paid by the scale above. In physical coordinates the words include all oriented face and corner saltation currents. The same quantities are limits of common-root finite differences under the cellwise quantile coupling.

Proof. Differentiate $\nu_a \mathcal{L}_a = \nu_a$ and $\nu_a(1) = 1$ on the strong-to-weak scale. The first derivative is

$$D\nu_a(G_a) = \nu_a(DG_a) + \nu_a((D\mathcal{L}_a)S_a Q_a G_a).$$

The Keller–Liverani resolvent identity on the scale just proved permits iteration. Each shape derivative is exactly the cellular current derivative of the moving partition, so no corner distribution is omitted. Normal convergence of the resolvent series and the finite number of incidence types justify termwise differentiation. The common-root coupling crosses the same oriented cells; its distributional derivatives are precisely the saltation currents, which proves the last assertion. $\square$

1.3. An explicit autonomous Hamiltonian realization. For each affine branch choose an exact symplectic isotopy $R_{i,\tau}$ from the identity to $(q, p) \mapsto (Q_i(q), P_i(p))$, supported in a Darboux collar assigned to branch i . The collars and clock intervals are pairwise disjoint. Let $K_{a,\tau}$ be the sum of the corresponding compactly supported Hamiltonians, periodically extended in $\tau \in \mathbb{R}/\mathbb{Z}$. On

$$\mathcal{M}_a = \widehat{\Sigma}_a \times T^*(\mathbb{R}/\mathbb{Z})$$

with symplectic form $\Omega_a = dq \wedge dp + dE \wedge d\tau$, set

$$H_a(q, p, \tau, E) = E + K_{a,\tau}(q, p).$$

The equation $\dot{\tau} = \partial_E H_a = 1$ fixes the return time exactly. At the end of a clock period the biseam recutting is the identity on physical coordinates and only changes the cell chart.

Theorem 1.4 (Autonomous biseam suspension). *The Hamiltonian flow of H_a has first return time one to the section $\tau = 0$ and first return map F_a . It is smooth on every corner stratum, its impact extensions agree on all oriented face and corner germs, and its Liouville flux on the section is $dq dp$. The construction is C^{m+1} in a on every compact parameter interval.*

Proof. Hamilton's equations on a branch collar are the prescribed exact isotopy, while outside the active clock interval they are the identity. Since the supports are disjoint, no compatibility equation couples two branch Hamiltonians. The biseam incidence rule identifies only equal oriented germs; the isotopies have matching flat jets at collar boundaries and hence paste on faces and corners. The equation $\dot{\tau} = 1$ gives return time one, and the stroboscopic map is the ordered branch isotopy followed by biseam recutting, namely F_a . Symplecticity gives the stated flux. Parameter regularity follows from the explicit affine generators and a relative Moser correction with fixed support. $\square$

1.4. Full-port work and calibration. Choose a clock interval preceding every branch isotopy and a smooth function ρ supported there with integral one. On the incoming vertical germ i put

$$\alpha_a = \epsilon_i \rho(\tau) d\tau.$$

The four incoming components are disjoint in the biseam completion, including all regular points and all oriented boundary germs.

Proposition 1.5 (Exact full-port work). *For every regular return segment entering through port i ,*

$$\int_0^1 \alpha_a(X_{H_a}) dt = \epsilon_i.$$

Consequently the mechanical work is the additive cocycle

$$W_n(x) = \sum_{j=0}^{n-1} \epsilon_{\iota(F_a^j x)}$$

on the full Liouville section. No positive-measure core restriction is used.

Proof. During the work interval $K_{a,\tau} = 0$, $\dot{\tau} = 1$, and the incoming port label is constant. Integration of $\epsilon_i \rho(\tau) d\tau$ gives ϵ_i . Concatenation gives the cocycle. $\square$

Theorem 1.6 (Conditional mechanical–valuation identification). *A smooth cash-additive law-invariant strongly time-consistent certainty equivalent generated by one utility is either conditional expectation or an entropic certainty equivalent with a constant coefficient ϑ . If, in the same work units, its normalized likelihood cocycle is required to equal the canonical mechanical likelihood generated by W_n , then $\vartheta = \theta$. Without this compatibility axiom mechanics does not identify an unrelated preference coefficient.*

Proof. Cash additivity yields the translation equation for the utility; differentiating twice gives constant absolute risk aversion. The tower property makes its coefficient time independent. Equality of the two normalized exponential likelihoods on the nonconstant full-port cocycle forces $(\vartheta - \theta)W_n$ to be constant, hence $\vartheta = \theta$. $\square$

Theorem 1.7 (Round-seven A1 closure). *The four-port benchmark possesses a genuinely bijective biseam symplectic return map, an explicit autonomous Hamiltonian suspension with exact return time, exact work on every regular incoming point, and finite-order response of arbitrary order on a polyhedral anisotropic current scale. The construction uses no isotropic Holder spectral-gap assertion.*

Proof. Combine Lemma 1.1, Theorems 1.2, 1.3, and 1.4, and Proposition 1.5. The calibration statement is Theorem 1.6. $\square$

SCOPE AND VERIFICATION STATUS

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